

ASUTOSH PANDA

📍 Brahmapur, India • ✉ asutosh.pda@gmail.com • 🌐 75asu.pages.dev • 🐙 github.com/75asu • 🔗 linkedin.com/in/75asu

SKILLS

GPU & AI Infrastructure: NVIDIA GPU Operator, DCGM, drivers/CUDA, MIG, vLLM inference serving, GPU SLOs and observability
Infrastructure & Networking: Kubernetes (GKE, bare-metal), Helm, Kustomize, FluxCD, ArgoCD, Terraform, OpenTofu, Ansible, Linkerd, KubeOne, OpenEBS, GCP, AWS, Hetzner, OVH, Cloudflare
Secrets and Auth: GCP Workload Identity Federation, External Secrets Operator (ESO), GCP Secret Manager, cert-manager, Pomerium
CI/CD and Observability: GitHub Actions, self-hosted ARC runners, Trivy, SonarQube, TruffleHog, Renovate; Prometheus, Grafana, Mimir, Loki, Tempo, Grafana Alloy, grafana-operator, Alertmanager
Development: Golang, Python, FastAPI, Bash, Node.js, TypeScript, PostgreSQL, Linux

PROFESSIONAL EXPERIENCE

ONE2N

Pune, Maharashtra, India

Senior SRE & Team Lead • Client: Fravity AI (Remote)

May 2026 - Present

- Built **fravity-spine**, the production multi-tenant GKE platform powering an AI product: private control plane, CMEK etcd, Workload Identity Federation, OpenTofu across **6** isolated state layers. Replaced bastions and shared SSH with identity-based **kubectrl** via GKE Connect Gateway + Workspace SSO; FluxCD-bootstrapped cert-manager, ESO, Traefik, and Pomerium SSO.
- Delivered the central LGTM observability platform: Mimir + Loki + Tempo (GCS-backed via Workload Identity, **365-day** metrics/logs + **30-day** traces, multi-tenant via X-Scope-OrgID) as authenticated public ingest. Migrated to grafana/alloy single-agent (logs, metrics, traces) across spine and the fravity-dev1 spoke; Alertmanager + Slack via ESO; grafana-operator CRDs for dashboards-as-code.
- Stood up a **5-tool** SAST stack on self-hosted ARC runners (Trivy operator, SonarQube CE, renovate-operator, TruffleHog) to meet the "scans inside our private GKE" audit bar. Shipped **scan-aggregator**, a FastAPI webhook service normalising **7** scanner outputs to a per-repo audit xlsx; first customer-facing deliverable: a customer security review with **76** H/C findings across **4** images.
- Audited GCP IAM/IaC/cost across **5** projects; surfaced **19** SA keys, an unowned **32-vCPU** instance, and **~\$26,520/yr** in zombie spend. Bootstrapped IaC for **2** additional envs to **60%** coverage; migrated **10** workload SAs + all human IAM to Terraform; rebuilt CI on Workload Identity Federation. Owned permanent-fix postmortems (flux_bootstrap_git race; Flux Kustomization circular-wait deadlock). Authored the Q2 SRE roadmap and lead a **2-engineer** team.
- Established the platform's first formal on-call and incident process: primary/secondary rotation, SEV1 response runbooks, and a weekly executive status cadence to the CEO and account owners.
- Delivered staging on **fravity-dev1** GKE as a commit-to-URL GitOps pipeline: CI build, Flux image-automation HelmRelease bumps, Atlas migration gate, a live TLS endpoint, and Slack deploy alerts, all **zero static credentials** via Workload Identity with per-app least-privilege GSAs. Root-caused the rollout's stacked failures (chart-default HPA runaway, cgroup OOMs, allowlist-bound DB networking) and upstreamed the chart fix.

LUXOR TECHNOLOGY

Seattle, WA, US

DevOps Engineer (Remote)

June 2023 - April 2026

- Operated a multi-cluster Kubernetes platform for **20+** teams and **3,000+** enterprise customers: **8+** clusters across Hetzner, OVH, and GCP with **24/7** uptime for L1/L2 validator and RPC workloads (BTC, LTC, Zcash, Stacks, tBTC), including **2** bare-metal HA clusters (KubeOne, OpenEBS) provisioned and driver-managed from scratch. Standardised on a **7-layer** FluxCD GitOps pipeline with Linkerd mTLS; migrated **70+** SOPS secrets to GCP Workload Identity.
- Led incident response for a Litecoin mainnet hard fork: cross-team build, **zero-downtime** rolling deployment across production, postmortem. Restored production after a kubelet TLS expiry on bare metal; deployed x509 monitoring across **10** clusters with **7-day/24-hour** PromQL alerts.
- Owned CI/CD for **50+** repositories on self-hosted ARC runners. Ported a failing Playwright E2E suite from Mac hosted runners (**\$5,000/mo**) to Linux ARC, cutting runtime **40 to 15 min** and cost to **\$600/mo**. Built code-signed Windows + Mac release pipelines and automated FluxCD deployments to GCP Artifact Registry.
- Built org-wide platform tooling: **KubeOrbit** (Go CLI for kubeconfig provisioning across **20+** clusters) and **Memphis** (Python SOC1/2/SOX compliance binary with auditor-ready Drive reports + Linear escalation; solo, **4 weeks** for the planned IPO). Prevented a supply-chain compromise by vendoring Trivy into a composite action across **9** repos; when CVE-2026-33634 later hit aquasecurity/trivy-action, Luxor pipelines were untouched.
- Proposed and built **BVT**, the org's first zero-hardcoded-secrets cluster: replaced SOPS-in-git with ESO + GCP Secret Manager + Workload Identity Federation; the pattern became the gold standard for new clusters. Authored custom Python tooling against Hetzner + OVH firewall APIs to enable secure cross-cloud peering for bare-metal nodes where no Terraform modules existed.

TATA CONSULTANCY SERVICES

Hyderabad, Telangana, India

Systems Engineer (Remote)

October 2020 - June 2023

- Migrated **2,500+** repositories from GitLab, Bitbucket, and SVN to GitHub Enterprise for **120** teams, saving **15 weeks** of manual effort.
- Built PowerShell, Python, and Golang automation across **22** RHEL/WebLogic servers (health checks and storage allocation), saving **380+ man-hours/month**.

PROJECTS

- Frit** (May 2026 - Present) 📄 - GPU SRE lab: vLLM on NVIDIA GPU Operator; OSS PRs to **dcgm-exporter** + **gpu-operator**.
- agent-box** (One2N, 2026) - Hardened devcontainer sandbox for AI agents: cap-drop ALL, non-root, no host credentials.
- Platform Zero** (Apr 2026 - Present) 📄 - **4** SRE labs (k8slab, ansiblelab, awslab, gcplab); **0-to-100** in one command.
- Truss** + **Dint** (2026) - Self-hostable BaaS (auth, storage, realtime) and an embeddable user-feedback widget.

TEACHING EXPERIENCE

- Full-Stack Instructor, **4** cohorts (2022-2026), taught NextJS, FastAPI, and DevOps to BSc CS students.

EDUCATION

SASTRA DEEMED UNIVERSITY

Master of Computer Applications (MCA) - Chennai, India - 2021 - 2023